

Effect of graphite oxide on the mechanical, thermal , structural and hydrophilicity proprieties of Polypropylene / Poly (Propylene-Glycol) Acrylate / Glycidyl Methacrylate PP/PPGA/GMA composite.

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ABSTRACT: This research aims to investigate the synergistic effects of grafting flexible fractions and incorporating reinforcing fillers in polypropylene (PP) matrix. The primary objective is to establish an experimental methodology for preparing and characterizing Polypropylene / Poly (Propylene-Glycol) Acrylate / Glycidyl Methacrylate (PP/PPGA/GMA) composites loaded with graphite oxide (GtO).

The process of oxidizing natural graphite powder using a mixture of sulfuric acid and nitric acid was validated through various analytical techniques including IRTF, AFM, DRX, XRF and ATG. The grafting reaction of PPGA and GMA on polypropylene and the incorporation of graphite oxide with varying levels were carried out in the melt state in a Brabender internal mixer; Dicumyl peroxide (DCP) was used as a radical initiator. A comprehensive assessment of the obtained materials was conducted through conventional analyses encompassing mechanical properties (tensile and impact), thermal behaviors (ATG and DSC), morphological characteristics (SEM) and the chemical state of film surfaces (Contact angle).

The investigation revealed two key outcomes. Firstly, it showcased the grafting effect of PP on the polymer's morphology and properties. Secondly, it underscored the consequential impact of incorporating graphite oxide on the overall behaviour of produced materials.

Keywords: graphite; graphite oxide; polypropylene; composite; grafting

1. Introduction

Polyolefins constitute a paramount class of plastics, with high consumption being entirely normal given their materials that provide an invigorating combination of physical and mechanical properties, as well as chemical stability. This makes them suitable for food and pharmaceutical applications [1]. Nevertheless, these properties remain valid under well-defined service conditions.

For instance, polypropylene (PP) serves as an illustrative member of this polyolefin family. Despite having several unique properties that meet industrial and commercial requirements, this polymer, at low temperatures, displays brittle behaviour and low impact resistance due to its high glass transition temperature, which is of approximately -10°C , and its relatively high crystallinity level [2].

In order to address these limitations, impact modifiers of a rubbery nature or thermoplastic elastomers (TPEs) are commonly employed. Ethylene-Propylene-Diene (EPDM) and Ethylene-Propylene copolymer (EPR) are the most common and effective choices due to their high impact resistance over a wide temperature range [3, 4, 5, 6]. However, the challenge arises regarding phase separation between PP matrix and elastomer fractions, despite both having a similar chemical structure [3, 6]. This issue stems from their incompatibility. Therefore, resorting to compatibilizing agents or surface treatments proves advantageous, as evident in studies where polar monomers like monomethyl itaconate (MMI), alkyl maleates, and maleic anhydride (MA) have been grafted onto polypropylene surfaces to prevent unstable morphology and poor interfacial adhesion [7, 8, 9, 10].

Another strategy involves the utilization of ethylene/alkyl acrylate copolymers as impact modifiers for PP, exhibiting effectiveness especially at lower temperatures. Furthermore, this approach offers the possibility of incorporating reinforcing fillers [11].

Her, identifying a filler that meets multiple criteria such as: easiness's dispersion, capacity to develop strong interactions with the matrix, ability to improve mechanical strength, thermal and structural stability, is a significant challenge.

Graphite emerges as a promising contender among potential fillers, boasting distinctive properties. It constitute of superposition of graphene sheets which consist exclusively of sp^2 hybridized carbon atoms, the graphene layers are linked by Van Der Waals forces, a unique amalgamation of metals and non-metals properties, its potentially lower cost than carbon nanotubes and graphene [12]. Graphite has attracted large attention for many industrial applications. Nonetheless, despite possessing these appealing properties, the challenge lies in the chemical inertness (lack of functional groups) and its limited specific surface area, making the dispersion of a significant quantity of graphite into polymer matrix a challenging task.

Graphite oxide (GtO) is produced by subjecting graphite to an extensive oxidation process using potent reagents like HNO_3 , H_2SO_4 , H_2O_2 , $KMnO_4$, $KClO_3$, and $NaNO_3$ [12, 13] , introducing oxygenated functional groups. These include epoxy and hydroxyl groups on the basal plane and carbonyl and carboxylic functional groups at the edges [12]. These reactive groups in GtO offer versatility for further modifications, augmenting its role as filler in polymer composites. Additionally, treating graphite chemically with concentrated sulfuric acid (H_2SO_4) and nitric acid (HNO_3) leads to the formation of exfoliated graphite oxide, promoting ready dispersion within the polymer matrix [14].

Exfoliated graphite oxide refers to graphite with a distinctive feature: a significant separation of adjacent carbon layers. This partial separation results in a cellular and porous microstructure that imparts an exceptional combination of elastomeric and viscous

characteristics. Additionally, this process leads to an expansion in volume and a reduction in density compared to non-exfoliated graphite. Furthermore, exfoliated graphite oxide also boasts an increased specific surface area [14, 15, 16].

Numerous polymer composites loaded with GtO have demonstrated advancements in mechanical, electrical properties, and thermal stability as substantiated by previous studies [17, 18, 19, 20, 21]. A notable enhancement in different proprieties was observed in the work of Kuvardina et al., where graphite oxide served as filler in a polypropylene matrix [22]. Moreover, the substantial impact of exfoliated graphite oxide on the crystalline structure of polypropylene nanocomposites was evidenced through experimental findings presented by Kyriaki et al [23].

The central focus of this study is to assess the influence of exfoliated graphite oxide, obtained through the oxidation of graphite using a concentrated mixture of sulfuric acid (H_2SO_4) and nitric acid (HNO_3), on the mechanical, thermal, and structural characteristics of PP/PPGA/GMA composites.

2. Experimental

Isotactic PPs with melt flow index (MFI) of 3.0 g/10 min (with a density of 0.905 g/cm³) were obtained from Repsol Quimica S.A. (Spain). Poly (Propylene-Glycol) Acrylate (PPGA), Glycidyl Methacrylate (GMA) and Dicumyl peroxide (DCP, 98%) were supplied from Sigma-Aldrich. Graphite powder particles (10 μm) were supplied from Qingdao Xinghe Graphite co. Ltd., China. The chemicals used for the synthesis of graphite oxide, sulfuric acid (H_2SO_4 , 97%), fumed nitric acid (HNO_3 , 69%) were obtained from Sigma-Aldrich without any further purification.

2.1. Synthesis of graphite oxide

A chemical treatment of natural graphite was used for the synthesis of graphite oxide (GtO): 5 g of purified powder graphite was dispersed in a mixture of concentrated sulfuric acid 200 ml and 140 ml of fuming nitric acid. The obtained mixture was ultrasonicated for 15 min to get an exfoliated GtO suspension then it was refluxed under a vigorous stirring at 130°C for 5 days.

Afterwards, 1000 ml of distilled water was added to the reaction vessel. The diluted slurry was then stirred for an additional 3 h. The product was filtered and washed with distilled water in order to remove impurities. The resulting GtO was then dried in a vacuum oven at 100 °C for 24 h and then grinded in fine particles.

2.2. Elaboration of composites

The various formulations and their respective designations used in this study are shown in Table 1. The grafting reaction and the preparation of composites were carried out in the molten state by using a 55 ml Brabender PL2100 mixer. PP, 0.1wt% of DCP, 1wt% of PPGA, and 1wt% of GMA were introduced in the preheated mixer at 180 °C and mechanically mixed at 50 rpm for 10 min. After six minutes of mixing, a certain amount of graphite oxide was added to the mixture.

The amount of graphite oxide added to each mixture was 0.2wt%, 0.4wt%, 0.6wt% and 0.8wt% respectively. The product was granulated and kept firmly away from the humidity in plastic bags.

[Table 1]

2.3. Characterization

2.3.1. ATR-FTIR spectroscopy

ATR-FTIR spectroscopy was used to confirm the formation and functionalization of GtO. About 50 mg of the GtO was compacted into thin pellets using a hydraulic press under a force

of 5 tons maintained for about 4 min. Samples were analyzed by signal averaging 16 scans at a resolution of 4 cm^{-1} using a JASCO-IR spectrometer.

FTIR analysis was also performed to confirm the grafting reaction of PPGA, and GMA onto PP. Thin films were obtained from the samples by compression molding between two Teflon sheets using a compression machine at $200\text{ }^{\circ}\text{C}$ and 180 kg/cm^2 . The spectra was recorded on a Shimadzu spectrometer in a transmission mode at 4 cm^{-1} with the coaddition of 32 scans with a spectral range of $4000\text{--}400\text{ cm}^{-1}$.

2.3.2. Atomic Force Microscopy (AFM)

AFM measurements were performed on a Bruker Dimension 3100V atomic force microscope, in Tapping (Dynamic) mode. The tips used have resonance of approximately 350 kHz.

2.3.3. Zeta-potentials analysis

Particle size distribution and zeta potential analyses were conducted using a dynamic light scattering at room temperature with the HORIBA Nano Partica SZ-100 instrument. This technique allowed for the determination of both the distribution of particle sizes and the magnitude of the particle surface charge.

2.3.4. X-ray diffraction (XRD)

Crystallographic structural analysis was conducted on both unaltered graphite and graphite oxide samples. This analysis involved the utilization of a Phillips System XPERT-PRO diffractometer, which operated with a $\text{CuK}\alpha$ X-ray source (wavelength $\lambda = 1.54\text{ \AA}$) at a generator voltage of 40 kV. The diffraction patterns, known as diffractograms, were obtained by scanning in the 2θ range spanning from 5 to 60° . This scanning process was carried out at a rate of 2° per minute.

For various composite samples, the X-ray diffraction patterns were obtained using a BRUCKER D8 diffractometer employing $\text{K}\alpha 1$ line from Cu ($1.54\text{ \AA}$), the operating setting

of the X-ray was 40 kV and 40 mA. The diffraction patterns were collected from 5 to 40° 2 θ at steps of 0.01° and intervals of 6 s per step.

2.3.5. X-ray fluorescence (XRF)

In order to ascertain the elemental composition and quantify the proportions of constituents present within distinct samples, both pristine graphite and graphite oxide, the analytical technique of X-ray fluorescence analysis (XRF) was employed. The samples were prepared as pellets and subjected to analysis using a ZSX Primus IV instrument. This instrument conducted the characterization under a vacuum environment at a temperature of 38°C, utilizing a power supply of 4.0 W and 30 kV for analysis in the semi-quantitative mode.

2.3.6. Thermogravimetric analysis (TGA)

TGA measurements studies of both GtO and their composites were performed using a Setaram SETSYS TG-DTA thermal analyser under nitrogen atmosphere at a flow rate of 20 cm³/min. About 10 to 20 mg of each sample was heated from 25 to 600 °C with a heating rate of 10 °C/min.

2.3.7. Differential Scanning Calorimetry analysis (DSC)

Crystallization (T_c) and melting (T_m) temperatures, crystallization (ΔH_c) and melting (ΔH_m) enthalpies are determined by NETZSCH DSC 200 F3, the sample amounts are about 10 mg and the experimental conditions used are as follows :

Samples are subjected to heat from 25 to 200 ° C on the first times to eliminate thermal history, then a descent from 200 ° C to 25 ° C, finally a second rise from 25 to 200 ° C. The heating rate is 20 ° C / min. The degree of crystallinity is calculated using the following equation:

$$\chi_c(\%) = \frac{\Delta H_m}{\Delta H_m^0 \cdot W_{PP}} \cdot 100$$

Where χ_c is the percentage of the crystallization, ΔH_m^{obs} is the measured enthalpy of melting, ΔH_m^0 is the theoretical crystallization enthalpy of the 100% crystalline PP, which has the value of 209 J g⁻¹ [32] and W_{PP} is the weight fraction of PP in the blend.

2.3.8. Mechanical properties

The tensile test was carried out on a universal mechanical test machine type Zwick. The test pieces are tested under normal conditions of temperature and pressure at a cross head speed of 20 mm/min. and according to ISO 527-1.

Notched Izod impact strength was carried out at room temperature on CEAST 9050 impact pendulum tester. The specimen prepared according to the specifications of ASTM D-256-88.

All measurements were made in five replicates and the values were averaged.

2.3.9. Scanning electron microscopy (SEM)

SEM observations were used to observe the oxidation effect on the graphite structure and to analyze the morphology and structural characteristics of the composites; also to verify the state of dispersion / distribution of the charges in the polymer matrix studied, a NEOSCOPE Scanning Electron Microscope was used with an acceleration voltage of 10 kV. Composites' specimens were frozen by immersion in liquid nitrogen to obtain brittle fracture.

2.3.10. Contact angle

Alterations in surface hydrophilicity of the distinct fabricated films were assessed by means of contact angle measurements employing sessile drop methodologies. The quantification of contact angles was executed utilizing a Ramé-Hart Model 590 F4 series goniometer and

tensiometer. All contact angle evaluations were conducted utilizing calibrated microsyringes to dispense 10 μL droplets of distilled water.

3. Results and discussion

3.1. ATR-FTIR analysis of the pristine graphite and graphite oxide

The oxidation process of graphite through treatment with a sulfuric and nitric acid mixture is substantiated by the emergence of distinct spectral bands. As illustrated by Figure 1, a distinct disparity between the spectra of pristine graphite and graphite oxide (GtO) is evident. The infrared spectroscopy (IRTF) analysis of pristine graphite (Figure 1a) reveals a solitary absorption band at 1640 cm^{-1} , characteristic of the vibrations related to the (C=C) bond within a conjugated system [24]. In contrast, the graphite oxide spectrum (Figure 1b) exhibits the presence of multiple absorption bands, ascribed to the various oxygenated functional groups located on the planar surfaces and edges of each graphene oxide sheet. The band situated at 3670 cm^{-1} corresponds to the elongation vibrations of hydroxyl (OH) groups stemming from carboxylic acids (COOH) and alcohols (OH). The pronounced band at 1720 cm^{-1} is associated with the elongation vibration of carbonyl (C=O) groups and carboxylic (COOH) groups, the band situated at 1362 cm^{-1} which refer to the stretching (OH) group of alcohol functions, while the band at 1210 cm^{-1} is attributed to ether (C-O-C) linkages [25]. These empirical findings substantiate the efficacy of acid mixture treatment, which effectively targets the exposed edges and structural irregularities within the graphite sheets.

[Figure 1]

3.2. Zeta-potentials and size particles analysis of the pristine graphite and graphite oxide

The zeta potential analysis of pristine graphite particles, as illustrated in Figure 2a, demonstrates a surface charge of moderately negative nature, approximately -20 mV . In contrast, graphite oxide particles exhibit a notably more negative surface charge, quantified at

around -33 mV (Figure 2b). This divergence in zeta potential values can be attributed to the introduction of electronegative functional groups onto the graphite framework during the oxidation process, as documented in reference [26]. These functional groups, primarily carboxylic (COOH) acid moieties, are capable of undergoing dissociation, leading to the formation of ($\text{COO}^- + \text{H}^+$) species at the particle's surface. This phenomenon contributes significantly to the augmentation of the zeta potential, thereby exerting a discernible influence on the stability of aqueous colloidal suspensions of graphite oxide, as discussed in reference [27].

[Figure 2]

These findings are supported by the images acquired from colloidal solutions containing pristine graphite dispersion and graphite oxide (Figure 3b) in distilled water. Once graphite was introduced to the aqueous medium, its particles exhibited a pronounced propensity for flocculation and aggregation. After a brief period (1 hour), these particles rapidly settled at the bottom of the container (Figure 3a). Conversely, the particles of GtO demonstrated an ability to establish more uniform and enduring colloidal dispersions (Figure 3b). These findings validate the effectiveness of the oxidation process, through which a significant number of oxygenated functional groups have been introduced onto the edges and the basal plane of GtO. These groups have attenuated the Van Der Waals interactions between the layers and imparted hydrophilicity, thereby greatly enhancing the hydration and aqueous exfoliation of these particles, especially with a brief sonication (few minutes) [28].

[Figure 3]

3.3. Morphological and size particles analysis of the pristine graphite and graphite oxide

By analyzing Figure 4, it becomes evident that the significant impact of the oxidation process on the resultant graphite oxide size can be inferred; as a diminution was observed at rate ten

times lower than the initial size of pristine graphite (10 μ m). This notable effect is attributed to experimental conditions. As proposed by Zhao et al. [29], the harsh conditions, under which the chemical oxidation process takes place, tend to rupture the σ bond of the carbonaceous skeleton. This phenomenon is consistently accompanied by the liberation of CO₂, resulting in the formation of graphene sheets with a smaller granulometry compared to the initial graphite.

[Figure 4]

The previous outcomes precisely corroborate what is affirmed by the AFM analysis imagery. Figure 5 displays 2D and 3D Atomic Force Microscopy (AFM) images of both pristine graphite (Figure 5a; Figure 5b) and graphite oxide (Figure 5c; Figure 5d). When comparing the images of these two analyzed samples, a significant difference in their appearance is evident, particularly marked by a noteworthy reduction in the thickness of the graphite oxide. These observed changes are probably attributed to the implantation and insertion of oxygenated functional groups along the edges and within the interlamellar spaces of the graphene sheets. Consequently, an exfoliate graphite oxide with sharp edges was produced (Figure 5d).

[Figure 5]

The acquired images via Scanning Electron Microscopy (SEM) are directed in the same orientation. As depicted in Figures 6, there is a notable contrast between the irregularly wide and flat pristine graphite sheets (Figure 6a ; 6b), which are closely interconnected, and the graphite oxide sheets (Figure 6c ; 6d), which exhibit a distinct flake-like shape, smaller dimensions, and a smoother surface. This distinctive morphology is likely a consequence of factors discussed previously, including the release of CO/CO₂ during the shock heating process and the intercalation of oxygen atoms between the graphene sheets during the oxidation process [29].

[Figure 6]

3.4. XRD analysis of the pristine graphite and graphite oxide

In Figure 7, the initial pristine graphite pattern reveals a prominent peak indexed as (002) and positioned at $2\theta = 26.3^\circ$, signifying an interplanar spacing around 0.33 nm. Post treatment, the distinct peak associated with graphitic structure diminishes, giving way to a novel peak at $2\theta = 24.1^\circ$, corresponding to an interlayer spacing approximately of 0.40 nm. This increase in separation between adjacent carbon layers is a direct outcome of integration of oxygenated functional groups introduced within the layered configuration of graphite oxide. This integration transpires through the catalytic action of compounds formed by the amalgamation of sulfuric and nitric acids. As a result, a separation of the graphene layers that make up graphite occurs, leading to the emergence of a new crystalline structure [30].

[Figure 7]

3.5. X-ray fluorescence analysis of the pristine graphite and graphite oxide

XRF spectroscopy stands out as one of the prominently employed techniques for both quantitative and qualitative assessment of the elemental composition of an analyte.

Through a brief examination of Figure 8, which displays the results obtained from XRF analysis, the noticeable disparities between the two curves become apparent. These discrepancies manifest both in terms of peak intensity and the angle of peak manifestation, reflecting variations in the composition of analyzed materials. Furthermore, these findings underscore a significant distinction among the characterized samples (pristine graphite and graphite oxide). The presence of oxygen as a prominent component, accounting for 19% of the graphite oxide powder, is attributed to the pronounced impact of the oxidation process that leads to the emergence of novel elements when compared to pristine graphite (Figure 8a) [31], where carbon is the predominant constituent with minor inclusions of Si, Ca and Fe.

[Figure 8]

3.6. TGA analysis of the pristine graphite and graphite oxide

Thermogravimetric analyses were conducted with the purpose of monitoring mass loss and temperature-associated thermal effects, aiming to acquire supplementary quantitative insights into the composition of distinct characterized samples (Figure 9). It is noteworthy that pristine graphite exhibits remarkable thermal stability (Figure 9a), as evidenced by the absence of any mass loss recorded until temperatures fell below 620°C.

[Figure 9]

In contrast, The TGA thermogram of graphite oxide (Figure 9b) reveals the existence of two distinct regions characterized by mass loss, both of which are directly associated with the oxidation process. The initial mass loss region, observed at approximately 80 °C, can be attributed to the evaporation of water molecules that are intercalated between the layers of graphene oxide, as reported by Wang et al. [32]. The second mass loss region, occurring around 250 °C, is linked to the decomposition of less stable oxygenated functional groups, specifically the C-O-C linkages and labile hydroxyl groups (OH). This decomposition releases a significant volume of gas in the form of carbon dioxide (CO₂), a phenomenon that can potentially contribute to the acceleration of the thermal degradation of the carbonaceous structure. This observation aligns with the findings of Wang et al. and Park et al. [32, 33].

3.7. Grafting reaction and elaboration of composites

Polypropylene (PP) underwent melt-phase functionalization through grafting with Poly(Propylene-Glycol) Acrylate (PPGA) and Glycidyl Methacrylate (GMA). Subsequently, graphite oxide (GtO) was introduced into the system utilizing a Brabender internal mixer. The grafting process involved a radical addition reaction initiated by alkoxyl radicals generated from the thermal breakdown of dicumyl peroxide (DCP). The phenyl radical formed is

particularly reactive because it is removed a hydrogen atoms generating a macroradical which has the initial active center of the grafting reaction. The macroradical derived from the abstraction of hydrogen adds the first side chain with polypropylene. PPGA chains are fixed on tertiary radicals.

The macroradical formed by the addition of a macromolecular chain of PPGA can then add the GMA monomers. But, the growing chain can also undergo transfer reactions with the hydrocarbon chain of polypropylene. These reactions stop the growth of the graft by generating an active center on another hydrocarbon chain, which can also initiate a new grafting reaction.

In fact, the initiates and propagation stages lead to a wide variety of radicals. The presence of PPGA and GMA lead to the formation of branched radicals which then have the possibility of reacting with graphite oxide rich in reactive oxygen groups.

The proposed reaction mechanism outlining the grafting process is delineated in (Figure 10).

[Figure 10]

3.8. FTIR analysis of grafted PP

In the context of this research paper, confirmation of the grafting of PPGA and GMA monomers onto the polypropylene substrate was achieved through a straightforward comparison between the IRTF spectra of neat PP (Figure 11a) and the grafted PP (Figure 11b). The IRTF spectrum of the grafted PP (PP/PPGA/GMA), as depicted in Figure 10b, exhibits distinct new peaks at 1730 and 1645 cm^{-1} , which are indicative of the presence of C=O carbonyl groups associated with the ester functions [34].

[Figure 11]

Additionally, a notable alteration in the two peaks at 1102 and 1167 cm^{-1} is observed in the grafted sample. Specifically, the intensity of the 1102 cm^{-1} peak, characteristic of C-O bonds,

is significantly heightened due to the increased concentration of these bonds resulting from the grafting reaction. Conversely, the intensity of the 1167 cm^{-1} peak, corresponding to C-H bonds in the PP matrix, diminishes due to the formation of new chemical bonds during the grafting process involving polar fractions rich in functional groups.

It is important to emphasize that the detection of characteristic bands related to GtO in the spectra of all loaded composites (Figure 11c; 11d; 11e and 11f) proved challenging, primarily due to the limited amount of GtO incorporated and its spectral overlap with the distinctive bands of grafted PP. This challenge parallels findings in a study by Barletta et al. [35], where they encountered similar difficulties in discerning GO-related bands in their investigation of Poly(lactic acid) reinforced with graphene oxide.

3.9. Mechanical properties of composites

3.9.1. Tensile

In our research study, we evaluated the impact of grafting and filler incorporation on the ultimate tensile properties, as characterized by stress and strain at break, as well as Young's modulus. Figure 12 illustrates a comparative analysis of these properties among neat polypropylene (PP), grafted PP (PP/PPGA/GMA), and composite systems (PP/PPGA/GMA/GtO) containing varying filler concentrations (0.2wt%, 0.4wt%, 0.6wt% and 0.8wt %).

[Figure 12]

Neat PP displayed typical hard and brittle material. In contrast, composite systems at different filler amounts exhibited ductile and resistant behaviour, attributed to the advantageous viscoelastic properties of graphite oxide cell walls [36, 37]. This effect became more pronounced with increasing filler content.

In the context of grafted PP, we observed a substantial 23.4% increase in elongation at break. However, tensile strength and Young's modulus experienced reductions of approximately 43.0% and 10.5%, respectively, compared to neat PP. This phenomenon can be linked to the random grafting of flexible moieties (PPGA/GMA) onto the well-ordered polypropylene chains, leading to a significant reduction in crystallinity.

Concerning the various composite systems investigated (PP/PPGA/GMA/GtO), our results revealed a significant enhancement in elongation at break, with the effect becoming more pronounced as the filler content increased. At 0.8wt% GtO, we observed a maximum improvement of 63.9%, indicative of a transition from brittle to ductile behavior. This transition can be attributed to the synergistic influence of grafting flexible components and the viscous properties of graphite oxide [38], when incorporated into the polymer matrix. Regarding Young's modulus in the composite systems compared with grafted PP, as expected, the addition of GtO resulted in a notable increase in this property, ranging from 1.8% to 9.6% and up to 10.2%, even at very low levels of filler incorporation (0.2wt%, 0.4wt% and 0.6wt% respectively). The possible reason for this increase is attributed to the solid nature of the added filler, characterized by its high Young's modulus, which influences the overall behavior of the reinforced matrix, especially when effective dispersion is achieved [39].

It is noteworthy that our findings align with prior research studies reporting similar trends in tensile properties [40, 41].

3.9.2. Impact resistance

The favorable conditions for robust interactions are established by the combination of small particle size, a wide array of functional groups, and the uniform dispersion of GtO filler within the PP/PPGA/GMA matrix enriched with polar groups. These interactions facilitate a continuous phase among the various constituents of the system.

As illustrated in Figure 13, the impact resistance of grafted PP exhibits a substantial increase of approximately 12% when compared to neat PP. This enhancement can be attributed to the introduction of flexible PPGA fractions, which possesses rubber nature and functions as a shock absorber. Furthermore, it is noteworthy that elevating the GtO content from 0.2wt% to 0.8wt% results in a significant improvement in impact strength for all composites, with a remarkable 27% increase observed in the system containing 0.8wt% GtO (compared to neat PP). This enhancement is linked in one hand, to the cellular structure of the exfoliated graphite oxide wall, displaying extraordinary elastomeric behavior [42]. In the other hand, to the efficient distribution of absorbed energy among the phases, facilitated by the well-dispersed charge within the polymer matrix and the strong adhesion provided by the functional groups of GtO and PPGA/GMA. Note that these results are in close agreement with what Kalaitzidou et al. [43] and Wakabayashi et al. [44] found when they reinforced polypropylene with graphite oxide.

[Figure 13]

3.10. TGA analysis of composites

In Figure 14, we present ATG thermograms of neat PP, grafted PP (PP/PPGA/GMA) and composites (PP/PPGA/GMA/GtO) with the different concentrations of GtO.

The grafting of flexible fractions onto the PP chains appears to exert a slight decrease in the thermal behavior (Figure 14b). In contrast, the PP/PPGA/GMA/GtO composites exhibit a relative improvement in thermal stability when compared to neat PP.

Upon closer examination of the thermograms Figure 14, it becomes evident that neat PP demonstrates no discernible volatile products below 400°C, with the principal PP chain decomposition occurring around 370°C. Conversely, all synthesized reinforced composites (PP/PPGA/GMA/GtO) consistently exhibit enhanced thermal stability relative to both pristine

PP and grafted PP (PP/PPGA/GMA). This increase suggests that the well-dispersed GtO particles, particularly prominent at lower concentrations, effectively serve as thermal barriers. These barriers impede the diffusion of volatile gases, such as CO and CO₂, which are well-known catalysts in the thermal decomposition process [45].

[Figure 14]

3.11. DSC analysis of composites

Key parameters such as the melting temperature (T_m), crystallization temperature (T_c), crystallinity content (X_c), and the heat of fusion (ΔH_m) for neat (PP), grafted PP (PP/PPGA/GMA), and their composites (PP/PPGA/GMA/GtO) within a temperature range of 80 to 220°C are presented in Tables 2.

DSC analysis reveals a consistent pattern across all materials, featuring a singular crystallization exotherm and a lone melting endotherm. These findings substantiate the presence of a singular crystalline population, underscoring the homogeneity of the system and the efficacy of the grafting process.

[Table 2]

Specifically, the T_m peak for neat PP is recorded at 166°C, but it noticeably decreases till 161°C for the PP/PPGA/GMA system. This reduction can likely be ascribed to the steric hindrance imposed by PPGA and GMA molecules on the PP chains, preventing them from coming into close contact and weakens molecular interactions and causing a decrease in crystallinity. This decrease is evident in the degree of crystallinity 42.1% compared to the 46.1% observed in neat PP.

Interestingly, the previous reduction in crystallinity observed in grafted PP (PP/PPGA/GMA) is offset proportionally by the introduction of graphite oxide, as indicated in Table 2. The highest X_c value of 48.6% is noted for the highest loaded composite PP/PPGA/GMA/0.8wt%

GtO, attributable to the nucleating effect of graphite oxide [23, 46]. especially, with its layered structure that can function as a scaffold for organizing polymer chains during the crystallization process. This structural characteristic orients the polymer chains to adopt a specific alignment, thereby exerting an influence on the resulting crystalline structure and enhancing nucleation efficiency. Notably, Kalaitzidou et al. [23] have suggested that this nucleation effect not only impacts the degree of crystallinity but also influences the crystal structures of the PP, resulting in thinner and more uniform crystals.

To emphasize, it is commonly agreed upon that any changes in the crystallization patterns of the polymer will be reflected in the mechanical and thermal properties of the composites. So, these findings may offer some insight into the overall behavior observed in characterized systems.

3.12. XRD analysis of composites

In the context of our scientific investigation, the X-ray diffraction pattern analysis of neat PP (Figure 15a) revealed the presence of four distinct peaks at 2θ values of 14.9° , 17.5° , 19.6° , and 23° , which have been associated with its typical crystalline structure [47, 48]. However, upon subjecting PP to a grafting process involving PPGA and GMA, notable alterations were observed in both position and intensity of these diffraction peaks (Figure 15b). This phenomenon aligns with findings reported by Doufnoune et al. [48, 49] who observed similar alterations when introducing ethylene propylene rubber (EPR) fractions onto polypropylene chains, attributing these changes to the modification of the PP crystallization process.

[Figure 15]

Notably, the introduction of even a low concentration of GtO resulted in the emergence of four additional peaks in XDR patterns (Figures 15c; 15d; 15e and 15f). Specifically, two of these peaks were observed at 2θ values of 16° and 21.1° , corresponding to monoclinic crystal

structures (α -form) , while the remaining two peaks appeared at 2θ values of 25.5° and 26.5° , indicative of trigonal crystal structures (β -form) [23]. It is worth highlighting that the intensity of these new peaks exhibited a positive correlation with the filler content, underscoring the potent nucleating effect of GtO.

3.13. Morphology Analysis

In this research study, we employed scanning electron microscopy (SEM) to scrutinize the fractured surfaces (cross-sections) of our composite specimens, with the fracturing process carried out in a cryogenic environment using liquid nitrogen. Figure 16 showcases SEM images obtained from the cryo-fractured surface of the pristine matrix. It is noteworthy that the fractured surface of the neat PP appears relatively smooth as it consists solely of homogeneous PP chains (Figure 16a). In contrast, when examining the grafted PP (Figure 16b), we observe a less smooth fractured surface characterized by prominent contrast, indicating that the grafting process causes a disruption in the crystalline structure and produces a kind of disorder in the PP morphology .

[Figure 16]

For further insights, we present SEM images in Figures 16c and 16d depicting composites incorporating mass fractions of 0.2wt% GtO and 0.4wt% GtO. The composites do not record any new morphological appearance with the absence of any trace of GtO, indicating an improved level of filler dispersion within the polymer matrix.

Conversely, Figures 16e and 16f showcase SEM images of composites containing 0.6wt% GtO and 0.8wt% GtO. These outcomes unequivocally underscore the pronounced influence of charge concentration on the dispersion status. These observations confirm the presence of charged particles uniformly dispersed and aligned within the matrix, likely attributed to the reduced spacing between them, as discussed in relevant literature [50].

3.14. Chemical state of film surfaces (surface hydrophobicity/hydrophilicity)

Through the measurement of the contact angle of tested films with water (Figure 17), the highest recorded contact angle is observed for neat PP (101.8°), indicative of its hydrophobic nature [51]. Additionally, it is noteworthy that the grafting process contributes to the reduction of hydrophobicity. A significant observation is also evident within the different composites, wherein contact angle decreases (96.6°, 93.2°, 89.3°, 89°) with increasing concentrations of GtO (0.2wt%, 0.4wt%, 0.6wt%, and 0.8wt%, respectively).

[Figure 17]

This trend suggests an enhanced hydrophilic character of the film surfaces. These findings are attributed to residual oxygenated groups present in the graphite oxide resulting from the oxidation process, which tend to diminish the surface retention of the grafted PP film, consequently leading to reduced interfacial energy with water [52]. This newly observed property of the composites holds particular significance when employed in food and sanitary product packaging, where sterilization and antibacterial capacity are paramount. In accordance with Bouchareb et al. [50], Bruinsma et al. [53], and Saulou et al [54], these results contribute, in one way or another, to imparting the final antibacterial property of the various prepared films, particularly as cell adhesion to solid surfaces hinge on material hydrophobicity/hydrophilicity.

Conclusions

Our research involved grafting PP with PPGA and GMA molecules, which was confirmed through IR spectroscopy. This grafting led to an increase in impact strength, along with a reduction in Young's modulus and crystallization rate. Investigating the impact of varying amounts of graphite oxide (GtO) on the mechanical, thermal, structural, and morphological properties using conventional methods revealed significant improvements in impact strength,

tensile resistance, and Young's modulus, even at low GtO loading rates. This enhancement can be attributed to the combined elastomeric and viscous behaviors of GtO, which is a non-polymeric material.

Furthermore, our findings demonstrated a notable increase in thermal stability alongside improved mechanical properties. The crystallization rate in composites with different GtO loadings exceeded that of grafted PP due to the nucleating effect of GtO. SEM micrographs verified the effectiveness of the grafting process and the proper dispersion of GtO within the polymer matrix, resulting in enhanced mechanical properties.

Additionally, our contact angle measurements indicated that GtO filler altered the chemical state of the films surface by increasing its hydrophilicity. These findings collectively provide valuable insights into the multifaceted effects of grafting and GtO loading on the properties of the composites studied in this research.

Declarations

Conflict of interest The authors declare no conflicts of interest /Competing interests regarding this article.

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